Fiscal Year:	FY 2007	**Task Last Updated:**	FY 02/01/2008
PI Name:	Hahn, Inseob Ph.D.		
Project Title:	Coexistence Boundary Experiment		
Division Name:	Physical Sciences		
Program/Discipline:	FUNDAMENTAL PHYSICS		
Program/Discipline--Element/Subdiscipline:	FUNDAMENTAL PHYSICS--Low temperature condensed matter physics		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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City:	Pasadena	**State:**	CA
Zip Code:	91109-8001	**Congressional District:**	27
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	00-HEDS-02
Start Date:	10/01/2000	**End Date:**	09/30/2007
No. of Post Docs:	0	**No. of PhD Degrees:**	
No. of PhD Candidates:	0	**No. of Master' Degrees:**	
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA JPL
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:	NOTE: Received NCE to 09/30/2007 per D. Strayer (1/07)		
Key Personnel Changes/Previous PI:	first		
COI Name (Institution):	Barmatz, Marty (Jet Propulsion Laboratory)		
Grant/Contract No.:	None		
Performance Goal No.:	1.24		
Performance Goal Text:	To explore the frontiers of physics and uncover mysteries of the universe obscured by Earth's gravity		
	In a simple fluid system, there is a strong gravity effect that leads directly to a distortion in the order parameter (density) near the liquid-gas critical point. Because of this large gravity-induced density gradient, ground-based experiments cannot be performed sufficiently close to the critical point to unambiguously test the scaling predictions and recent theoretical models of the equation-of-state. Calculations have shown that in a microgravity environment the coexistence curve can be attained within two additional decades in reduced temperature of at the 3He liquid-gas critical point. The basic goal of this experiment is to measure the coexistence curve near the 3He critical point in the low gravity environment of space. The coexistence curve leading and correction amplitudes as well as critical exponent will be obtained from the measurements. This study will be performed together with other thermodynamic properties being planned for the MISTE project. Combining results of these simultaneous measurements will provide a unique		

Task Description:	opportunity to unambiguously test results of the scaling hypothesis and recent crossover equation-of-state models. The main objective of the COEX (Coexistence Curve Experiment) is to obtain the coexistence curve of 3He near the liquid-gas critical point using the same hardware designed for the MISTE experiment. The specific scientific objectives of this experiment are: 1) To measure the liquid-vapor coexistence curve of 3He in the range $0.01 < \|r-rc\|/rc < 0.1$ in a microgravity environment and test recent crossover equation-of-state models. 2) To obtain leading and correction amplitudes and to compare universal ratios with the recent theoretical calculations. 3) Test the scaling hypothesis prediction with results from the MISTE experiment. 4) Obtain the rectilinear diameter from the coexistence curve measurements and test theoretical prediction for the deviation from "the law of rectilinear diameter." Precise experimental validation of the scaling law and test of modern equation-of-state models by combining results from many different ground-based experiment typically yields not reliable or accurate enough results. This is due to the fact that it is extremely difficult, if not impossible, to find and eliminate inconsistencies in the experimental conditions and in the procedures adopted for data analysis in the different experiments. Furthermore, gravity effects have always been a dominant factor in preventing experimental measurements from being obtained in the asymptotic region. This task combining with the current MISTE flight project will give us unique experimental opportunity to perform the systematic test of the scaling law hypothesis and equation-of-state models.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	A key feature of a continuous phase transition in many physical systems in Nature is scale-invariance in fluctuations of the order parameter, which implies continuity between large and small scale fluctuations associated with the transition. Scale-invariance occurs in many systems that are still not completely understood, such as the universe at the onset of the big bang as currently envisioned by cosmologist, the clustering of galaxies, massless fields in elementary particle physics, the distribution of earthquakes, and turbulence in fluids and plasmas. In each of these examples there is a wide range of scales over which some phenomena vary as a power law of the scale. The Nobel prize winning renormalization-group theory is the most successful approach thus far developed to derive and explore the relations between the characteristics of a system as viewed on different length scales. The Microgravity Scaling Theory Experiment (MISTE) and Coexistence Curve Experiment (COEX) will test the theory with unprecedented accuracy.
Task Progress:	The shape of the liquid-gas coexistence curve of 3He very near the critical point temperature Tc was measured using the quasistatic thermogram method. This study was performed in the Earth's gravitational field using two cells of very different heights (0.5 mm and 48 mm). The measured coexistence curve near the critical point was strongly affected by the gravitational field. Away from the critical point, we compare the coexistence curve obtained using the thermogram method with earlier work by Pittman et al. The recently developed crossover parametric model of the equation-of-state is used to take gravity effects into account. The shape of the measured coexistence curve very near the critical point is remarkably symmetric about the critical density. Our results close to the critical point are consistent with the slope of the rectilinear diameter obtained by Pitman et al. from measurement father away from Tc. The deviation from a law of rectilinear diameter predicted by revised scaling and the Yang-Yang anomaly were not observed in 3He within the 0.1% accuracy in our measurements. [Ed. note: submitted September 2011 when report became apparent.]
Bibliography Type:	Description: (Last Updated: 01/02/2025)
Articles in Peer-reviewed Journals	Barmatz M, Hahn I, Lipa JA, Duncan RV. "Critical phenomena in microgravity: Past, present, and future." Reviews of Modern Physics, 2007 Jan-Mar;79(1):1-52. http://dx.doi.org/10.1103/RevModPhys.79.1 , Jan-2007